

March 18, 2025

Email to: IRPDataRequest@cpuc.ca.gov

Subject: VGIC's Informal Comments regarding the Draft 2022-2023 Integrated Resource Planning ("IRP") Inputs and Assumptions ("I&A")

Re: Informal Comments of the Vehicle Grid Integration Council Regarding the Draft 2024-2026 Integrated Resource Planning Inputs and Assumptions

The Vehicle Grid Integration Council ("VGIC") appreciates the opportunity to comment on the California Public Utilities Commission's ("CPUC") Draft 2024-2026 Integrated Resource Planning ("IRP") Inputs and Assumptions ("I&A"). VGIC appreciates the CPUC and Energy Division ("ED") staff's commitment and efforts to engage stakeholders on this critical initiative.

VGIC is a 501(c)6 membership-based advocacy group committed to advancing the role of electric vehicles ("EV") and vehicle-grid integration ("VGI") through policy development, education, outreach, and research. VGIC supports the transition to a decarbonized transportation and electric sector by ensuring the value from flexible EV charging and discharging is recognized and compensated to achieve a more reliable, affordable, and efficient electric grid. VGIC has actively engaged in policy and research efforts to understand the need and opportunity for VGI better. As such, we offer the below recommendations for ED staff's consideration.

I. INTRODUCTION & SUMMARY.

VGIC appreciates the CPUC's continued efforts to improve the modeling of VGI and demand flexibility from EVs within the IRP. EVs are unlike any other modifiable load or dispatchable resource type because individual EVs represent a more significant potential load shift and energy storage capacity than the typical customer-sited distributed energy resource ("DER"). Moreover, the deployment of EVs and flexible EV charging and discharging solutions is expected to grow at an extraordinary rate due to underlying policy drivers and regulations like the California Air Resources Board's ("CARB") Advanced Clean Cars II regulation and, potentially, the implementation of Senate Bill 59, which may require certain vehicle classes sold in California to be bidirectional-capable by an as-yet-to-be-determined date. As a result, the statewide fleet of deployed EVs and installed EV chargers will offer immense latent load flexibility and energy storage capacity that may eclipse availability of other DERs, even at very modest EV customer participation rates.

While VGIC commends the Commission for considering the capabilities of EV charging equipment within the IRP modeling effort, there have not been many substantial changes between this version of the IRP I&As and last cycle's inputs. Overall, VGIC continues to recommend the ED and E3 staff take additional steps to better model VGI within the IRP. We offer the following comments:

- The CPUC should release the results of VGI modeling scenarios.
- The CPUC should work towards adding VGI as a full candidate resource in the IRP.
- VGIC recommends modeling increased V2G potential, including from electric school buses.

II. COMMENTS.

A. The CPUC should release the results of VGI modeling scenarios.

VGIC was encouraged by the inclusion of VGI in the I&As for the last IRP cycle. However, VGIC has been unable to find the results of any analysis that was done last cycle that included VGI resources. VGIC is unclear if the sensitivity analyses, including the proposed Mid-VGI and High-VGI scenarios were ever executed by the E3 team. If they were, it is unclear where the results have been posted. Since distributed energy resources (DERs) and EV load flexibility is relatively new within the IRP process, many DER and VGI stakeholders are less familiar with the RESOLVE model. It is difficult to parse through the available RESOLVE spreadsheets to find different results that are not discussed or presented by Energy Division in venues such as Rulings or workshops. To ensure that all stakeholders can understand the results of different sensitivity analyses, VGIC encourages the CPUC to include the results of any VGI analyses in published materials and workshops that discuss modeling outputs. A dedicated VGI or load flexibility workshop may also be warranted to ensure there is enough time for discussion on these topics. While we acknowledge VGI is not the highest priority among the many IRP objectives, even high-level results and dedicated discussion related to modeling VGI would go a long way in filling a critical knowledge gap for VGI stakeholders, including policymakers and customer groups. VGIC has heard CPUC and CEC Commissioners, as well as senior CAISO representatives, in both public settings and private/ex parte meetings ask for more details on the value of VGI on California's grid. We believe this IRP cycle is an important opportunity to answer these Commissioner's questions.

B. The CPUC should work towards adding VGI as a full candidate resource in the IRP.

VGIC encourages the CPUC to incorporate VGI as a full candidate resource in the IRP as soon as possible. There is significant potential for VGI to contribute to California's electric system and provide ratepayer savings. While some modeling has been done on more limited applications of VGI and exploration may soon begin on the ability of VGI to greatly improve distribution system utilization, California has yet to evaluate VGI as a tool in electric system portfolio development. Without explicit consideration of VGI potential in system planning, California risks planning for, and ultimately procuring, excess generation that is not needed due to VGI contributions. This is especially important given the amount of EV charging load that is included in California's demand forecast that is used for the IRP.

Significant amounts of work have already been done through these I&As to gather cost data for VGI resources, develop adoption curves, and create total resource potential estimates. If additional work or data is needed to fully model VGI in the core capacity expansion analysis, Energy Division should share what information would allow for full inclusion in the IRP. VGIC understands the complexity of accurately incorporating all aspects of VGI into an IRP model. We would welcome collaboration with Energy Division to continue strengthening VGI representation so that California can evaluate all tools in its toolkit.

C. VGIC reiterates its recommendation to model increased V2G potential, including from electric school buses.

The 2025 Draft I&As continue to assume that “V2G potential starts at 0% of V1G potential in 2025 and grows to 50% of V1G potential in 2050.”¹ This underestimates the amount of V2G that could be available in California, especially considering the current directive for the CEC, in consultation with the CPUC and CARB, to implement SB 59, which may require certain vehicle classes be bidirectional capable by a certain year. With this major development in mind, VGIC respectfully requests that the IRP modeling team conduct one or more sensitivity analyses to model higher values of V2G potential. This will be an incredibly meaningful effort to inform ongoing CPUC and CEC initiatives aiming to better understand V2G resource potential.

VGIC believes it is reasonable to model at least one sensitivity for higher V2G potential based on three incorrect assumptions that are still in the I&As. Firstly, the I&As state that “technology readiness” for V2G is needed, when V2G systems have already been deployed in California and the technology readiness level (“TRL”) of V2G is currently a 9: *Actual System Proven in Operational Environment*. In fact, the largest V2G site in North America is located in Oakland, California, and provides peak reduction services to Pacific Gas & Electric. Secondly, the updated I&A document still insinuates that FERC Order 2222 or some other overarching policy framework is needed for V2G deployment. While FERC Order 222 may support market development for certain aggregators in the long-term, the primary drivers of V2G compensation mechanisms and site development include the Emergency Load Reduction Program (ELRP), the CEC’s Demand Side Grid Support Program (DSGS), PG&E’s V2X Pilots, and SDG&E’s Dynamic Export Rate Pilot all available to customers today. Additionally, SCE has proposed a V2G Export Credit (“Vehicle-to-Grid Resource Proposal”) in its recent General Rate Case Phase II. Overall, V2G is a proven technology that is already deployed and receiving compensation in California.

Third, the updated inputs and assumptions document states the 50% in 2050 “is entirely for planning purposes, considering that not all OEMs are willing to enable vehicles to be V2G capable and warranty battery for grid use by 2050 and not all drivers will want to use their vehicles as a grid asset.” Today, more than 50% of OEMs, representing *far greater* than 50% of California EV sales, have publicly stated they will enable vehicles to be V2G capable. Light-duty vehicle OEMs that already offer bidirectional charging capability include: Nissan, Ford, GM (including Chevrolet, Cadillac, and GMC), Kia, and Tesla.

¹ Draft I&As at p.107.

Additional manufacturers have made public statements clarifying their intent to offer V2G solutions, including Hyundai, Volkswagen, BMW, Mercedes-Benz, Lucid, Audi, Volvo, Polestar, and Rivian. In the competitive US automotive market, public statements made by major automakers regarding future vehicle capabilities are not made lightly. Regarding the second statement that not all drivers will want to use their vehicles as a grid asset, this is already accounted for in the V1G potential curve.

Additionally, VGIC reiterates its recommendation to incorporate electric school buses (“ESBs”) in IRP modeling, given that ESBs represent an ideal Vehicle-to-Grid (“V2G”) use case due to the predictable and consistent alignment between ESB dwell times and summer system peaks. The interest in V2G ESBs has only grown since the last IRP cycle, with the World Resources Institute (“WRI”) sharing that California has over 3,110 committed electric buses across the state, of which nearly 45% are delivered or operating.² A majority of these ESBs will be Type C, which come equipped with between 126 kWh and 315 kWh of battery capacity.³ Moreover, all ESBs claiming credits under CARB’s HVIP or ZESBI programs, which represent the vast majority of California’s ESB, are **currently required to be capable of bidirectional charging**. For load and generation profiles for the IRP modeling effort, VGIC recommends leveraging load and generation profiles from existing ESB deployments, including managed charging and bidirectional charging pilots such as those implemented by PG&E and SDG&E. Overall, VGIC believes the exceptional anticipated growth in latent battery capacity from ESBs and the significant alignment between dwell time and system peak merits consideration in the IRP modeling effort.

With this in mind, VGIC respectfully requests the IRP modeling team conduct a “V2G_High” sensitivity analysis that assumes a ramp-up in V2G potential as a percentage of V1G potential from 0% in 2025 to **90% in 2050**.

III. CONCLUSION.

VGIC appreciates the opportunity to provide these informal comments on the Draft 2024-2026 IRP Inputs & Assumptions. We look forward to further collaboration with the Energy Division and other stakeholders in this docket.

Respectfully submitted,

/s/ Zach Woogen

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VEHICLE-GRID INTEGRATION COUNCIL

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² Lydia Freehafer, Leah Lazer, and Brian Zepka. *The State of Electric School Bus Adoption in the US*. Updated November 18, 2024. <https://www.wri.org/insights/where-electric-school-buses-us>.

³ Jessica Wang, Vishant Kothari, Emmett Werthmann, Eleanor Jackson, Lidia Henderson, Alissa Huntington. Updated August 2024. “Electric School Bus Market Study and Electric School Bus U.S. Buyer's Guide” Washington, DC: World Resources Institute. <https://electricschoolbusinitiative.org/electric-school-bus-market-study-and-electric-school-bus-us-buyers-guide>